Problem1

(a) Violation type: Foreign Key. The SuperSSN '777553333' does not exist in the EMPLOYEE table's SSN column.
(b) Violation type: Entity Integrity. The primary key of Dept_Locations is (Dnumber, Dlocation). Neither part can be null.
(c) Violation type: Primary Key. Dnumber '1' already exists in DEPARTMENT. Primary keys must be unique.
(d) Violation type: Entity Integrity. The primary key of WORKS_ON is (ESSN, Pno). Pno cannot be null.
(e) Violation type: Primary Key. The pair ('453453453', 'Michael') already exists in DEPENDENT.
(f) Violation type: Foreign Key. Dnumber '5' is referenced by EMPLOYEE (Dno), DEPT-LOCATIONS(Dnumber) and COMP_PROJECT (Dnumber). Deleting it causes "orphaned" records.
(g) Violation type: Foreign Key. SSN '999887777' is referenced by WORKS_ON (ESSN). Deleting the employee violates referential integrity.
(h) Violation type: Foreign Key. 'ProductX' (Pnumber 1 and 30) is referenced in WORKS_ON. Deleting the project leaves WORKS_ON records with no parent.
(i) Violation type: Foreign Key. Changing Dnumber from 5 to 7 violates integrity because existing employees and projects reference number 5.
(j) Violation type: Foreign Key. The new SuperSSN '345678912' does not exist in the EMPLOYEE table.
(k) Violation type: None. This is a valid update to a non-key attribute (Hours).

Problem2

(a)
$\rho(PX,\sigma_{Pname='ProductX'}(Com\_Project))$
$\rho(WX,\sigma_{Hours>5}((PX)\bowtie Works\_on))$
$EmpX = \pi_{ESSN} (WX)$
$\rho(PY,\sigma_{Pname='ProductY'}(Com\_Project))$
$\rho(WY,\sigma_{Hours>5}((PY)\bowtie Works\_on))$
$EmpY = \pi_{ESSN} (WY)$
$Both=EmpX \cap EmpY$
$\rho(Dept4,\sigma_{Dno=4}(Employee))$
$Result \leftarrow \pi_{SSN, Fname, Lname}(\rho_{SSN \leftarrow ESSN}(Both) \bowtie Dept4)$

(b)
$\rho(T1,\pi_{ESSN}\ \rho(D,Dependent))$
$\rho(T2,\pi_{SSN}\ \rho(E,Employee))$
$Result \leftarrow \pi_{SSN, Fname, Lname}(E\bowtie(T2-T1))$

(c)
$\rho(J, \sigma_{Fname='James' \wedge Lname='Borg'}(\rho(E,Employee)))$

ρ(T1, π_{SSN}(σ_{SuperSSN = J.SSN}(E)))
ρ(T2,π_{E2.SSN}(ρ(E2,Employee) ⋈_{E2.SuperSSN = T1.SSN} T1))
Result ←π_{SSN, Fname, Lname}(Employee⋈T2)

(d)
ρ(E, Employee)
ρ(W, Works_on)
ρ(T, π_{ESSN}(W))
Result ← π_{SSN, Fname, Lname}(E ⋈_{E.SSN = T.ESSN} T)

(e)
ρ(P, Com_Project)
ρ(W, Works_on)
ρ(S1, π_{Pnumber}(σ_{Plocation='Stafford'}(P)))
ρ(S, ρ_{Pnumber → Pno}(S1))
ρ(R, π_{ESSN, Pno}(W))
ρ(Ans, R/S)
Result ← π_{SSN, Fname, Lname}(Employee ⋈_{Employee.SSN = Ans.ESSN} Ans)

(f)
ρ(D, Department)
ρ(E, Employee)
ρ(MaleEmp, π_{SSN,Salary}(σ_{Dno='5' ∧ Sex='M'}(E)))
ρ(E1, MaleEmp)
ρ(E2, MaleEmp)
ρ(NotMin, π_{E1.SSN,E1.Salary}(σ_{E1.Salary > E2.Salary}(E1 × E2)))
ρ(MinEmp, MaleEmp − NotMin)
ρ(Dept5,σ _{Dnumber='5'}(D))
Result ←π _{Dname,Salary}(Dept5 ⋈_{Dept5.Dnumber='5'} MinEmp)

(g)
ρ(Dept4Emp,π_{SSN, Fname, Lname, Salary} (σ_{Dno=4}(ρ(E, Employee))))
ρ(E1, Dept4Emp)
ρ(E2, Dept4Emp)
ρ(NotMax,π_{E1.SSN, E1.Fname, E1.Lname, E1.Salary} (σ_{E1.Salary < E2.Salary}(E1 × E2)))
ρ(MaxEmp,Dept4Emp − NotMax)
Result ←π _{SSN, Fname, Lname}(MaxEmp)

(h)
ρ(E, Employee)
ρ(P, Com_Project)
ρ(W, Works_on)
ρ(DL, Dept_Locations)
ρ(HouProj,π_{Pnumber}(σ_{Plocation='Houston'}(P)))

ρ(EmpOnHouProj,π_{ESSN}(W ⋈_{W.Pno = HouProj.Pnumber} HouProj))
ρ(EmpOnHouProj2,ρ_{ESSN → SSN}(EmpOnHouProj))
ρ(HouDept,π_{Dnumber}(σ_{Dlocation='Houston'}(DL)))
ρ(EmpHouDept,π_{SSN}(E ⋈_{E.Dno = HouDept.Dnumber} HouDept))
ρ(AnsSSN,EmpOnHouProj2 ∩ EmpHouDept)
Result ← π_{SSN, Lname, Fname, Address}(E ⋈_{E.SSN = AnsSSN.SSN} AnsSSN)

(i)
ρ(E, Employee)
ρ(D, Department)
ρ(Dep, Dependent)
ρ(Managers, π_{MgrSSN}(D))
ρ(SpouseEmp, π_{ESSN}(σ_{Relationship='Spouse'}(Dep)))
ρ(SpouseEmp2, ρ_{ESSN → SSN}(SpouseEmp))
ρ(AnsSSN, Managers ∩ ρ_{MgrSSN → SSN}(SpouseEmp2))
Result ← π_{SSN,Lname}(E ⋈ AnsSSN)